Reductant + Oxidant $\rightleftharpoons$ Oxidized reductant + reduced oxidant

Example : oxidation of copper by iron

$$Cu^{+} + Fe^{3+} \rightleftharpoons Cu^{2+} + Fe^{2+}$$

Electrons cannot be created or destroyed in a chemical reaction and therefore if one molecule is oxidized the other one must be reduced, known as a redox reaction. The complete oxidation of glucose by oxygen is described in the following reaction.

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O \quad \Delta G^{\circ\prime} = -2823 \text{ kJ.mol}^{-1}$$

The above reaction can be divided into two half reactions. In the first reaction, the carbon atoms of glucose are oxidized as follows:

$$C_6H_{12}O_6 + 6H_2O \longrightarrow 6CO_2 + 24H^{+} + 24e^{-}$$

In the second half reaction, molecular oxygen is reduced

$$6O_2 + 24H^{+} + 24e^{-} \longrightarrow 12H_2O$$

In living organisms, these half reactions occur through coordinated multistep pathways that harness the liberated free energy in the form of ATP. In the above reaction, the electrons involved are not transferred directly to oxygen, whereas the transfer process occurs through coenzymes NAD^{+} and FAD to form 10NADH and $2FADH_2$ through several glycolytic enzymes such as glyceraldehyde 3 phosphate dehydrogenase, pyruvate dehydrogenase and citric acid cycle enzymes – isocitrate dehydrogenase, α-ketoglutarate dehydrogenase, succinate dehydrogenase and malate dehydrogenase. The stepwise passage of electrons from NADH and $FADH_2$ to O_2 through a series of electron carriers is coupled to the translocation of protons from the mitochondrial matrix to the intermembrane space. The free energy stored in the resulting pH gradient drives the synthesis of ATP from ADP and Pi through oxidative phosphorylation.

8.1.1 Redox potential

The oxidation reduction potential, E, in general, redox potential, is a measure of the affinity of a substance for electrons and is measured relative to hydrogen. In principle, a positive redox potential implies that the substance has higher affinity for electrons than does hydrogen and is anticipated to accept electrons from hydrogen. In negative redox potential, the substance has lower affinity for electrons than hydrogen and therefore would donate electrons to H^{+}, forming hydrogen. Any atom or molecule, which is more electronegative than hydrogen is assigned a positive (+) redox potential; those less electronegative a negative (–) redox potential. Greater the difference between the redox potentials of two substances (ΔE), greater the electrons flow spontaneously from the less positive to the more positive side. ΔE is expressed in volts.

XII U8 Biological oxidation.indd 189 4/8/2020 3:25:55 PM